

- 1 (a) (i) radial lines..... B1
 pointing inwards B1
- (ii) no difference OR lines closer near surface of smaller sphere B1 [3]
- (b) (i) $F_G = GMm/R^2$ C1
 $= (6.67 \times 10^{-11} \times 5.98 \times 10^{24}) / (6380 \times 10^3)^2$
 $= 9.80 \text{ N}$ A1
- (ii) $F_C = mR\omega^2$ C1
 $\omega = 2\pi/T$ C1
 $F_C = (4\pi^2 \times 6380 \times 10^3) / (8.64 \times 10^4)^2$
 $= 0.0337 \text{ N}$ A1
- (iii) $F_G - F_C = 9.77 \text{ N}$ A1 [6]
- (c) because acceleration (of free fall) is (resultant) force per unit
 mass B1
 acceleration = 9.77 m s^{-2} B1 [2]
- 2 (a) (i) a , ω and x identified (-1 each error or omission) B2